How varied are stellar populations in the Solar vicinity? Exploring GCNS and GALAH DR4 with t-SNE

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Abstract

We select a set of stars in the Solar vicinity observed by the Gaia Catalogue of Nearby Stars (GCNS; which comprises approximately 330k stars within 100 pc of the Sun) [2] that have also been followed-up with high-resolution spectroscopy by the GALactic Archaeology with HERMES (GALAH) DR4 survey [1]. In this context, the visualization of multidimensional data with potentially non-linear relationships remains a significant challenge. Our goal is to verify whether t-distributed Stochastic Neighbor Embedding (t-SNE) can retrieve the distinct stellar population properties in an unsupervised manner (effectively by clustering similar stars together), and to explore whether peculiar stars, such as emission-line objects, stand apart in the map (outlier detection itself is a later step). Hence, our study focuses on exploring how the t-SNE captures the main spectral features of the ∼5,000 stars in our sample. We tested and validated the t-SNE projection through spectral parameter-based color-mapping, which revealed a smooth gradient across the projected data, demonstrating that the algorithm successfully preserves the intrinsic physical correlations between several stellar parameters, such as $T_{\rm eff}$, $\log g$, and [Fe/H]. Furthermore, we examined the perplexity hyperparameter as a weighting factor between manifold scales, noting that a value of 30 increases sensitivity to local structures, whereas a value of 50 prioritizes the preservation of the global topology. Our preliminary results indicate the presence of a small detached sub-cluster, which maintains a different internal parameter gradient, distinguishing it from the main stellar group as a unique population within the local manifold, which requires further investigation.

1. Introduction

The cross-match between GCNS [2] and GALAH DR4 [1] within 100 pc was characterised at catalogue level in our SAB25 proceedings [8]:

6,198	FGK	1.5 Gyr	−0.19 dex	Thick disk
common to GCNS & GALAH DR4	dwarf dominated	median age	median [Fe/H]	plus probable halo

High-resolution spectra are high-dimensional and carry subtle, non-linear variations that linear projections blur. Manifold learning embeds them in two dimensions directly from flux morphology, with no synthetic grids or labels; the methods are defined in Section 2.

2. Data and Methodology

Step	DATA	MAP	FILTER	GROUP	CHECK	What happens
1 Sample	6,198					GCNS × GALAH DR4: **6,198 stars** within 100 pc [1,2]; FGK dwarfs dominate.
2 Spectra	5,304					4 CCDs, native pixels, **14,245 px** per star; 894 lack a full set, so **5,304 mapped**.
3 Projection		Fig. 1a				On the 5,304 stars with all 4 CCDs: cosine metric, no PCA [7], no z-score; ***t*-SNE** [3,6] (perplexity 30) and **UMAP** [4].
4 Colour, flag			Fig. 1b, 2			Cuts ($T_{\rm eff} \geq 4500$ K, S/N ≥ 10 in 4 CCDs) and flags only filter the display; $T_{\rm eff}$, $\log g$, [Fe/H] only colour the map.
5 Emission				Fig. 3		**HDBSCAN** [5] on the 286 emission-flagged stars: dense groups of any shape, subclusters S0–S4.
6 Settings					Fig. 4	Same stars under *t*-SNE perplexity 30, 50 and UMAP 15, 30: overlap **0.21–0.37**, distance correlation **0.38–0.49**.
7 Spectra check					Fig. 5	Emitters vs. quiet dwarfs, line by line: the **H**α line (6563 Å) tests the flag.

PCA vs. *t*-SNE & UMAP
PCA [7] is linear and blurs subtle differences (not applied); *t*-SNE [3,6] and UMAP [4] are non-linear and keep similar spectra together.

Perplexity
Neighbours each star "looks at": 30 favours local structure, 50 the global topology.

Overlap & distance corr.
Overlap: share of the 15 nearest neighbours kept in 2D (local). Distance correlation: Spearman of pairwise distances (global).

HDBSCAN
Dense groups of any shape, no preset number; stars outside any dense group stay unassigned.

3. Results

Figure 1 — Main unsupervised projection (*t*-SNE, perplexity 30; 5,304 stars). **(a)** Coloured by effective temperature ($T_{\rm eff}$); markers: emission (Hα), SB1/SB2 (binary candidates) and low S/N. One point is one star; close points are similar spectra; axes have no units. **(b)** Same map coloured by $T_{\rm eff}$, $\log g$ and [Fe/H]. These properties never enter the algorithm; they only test whether spectral similarity alone groups stars by atmospheric properties.

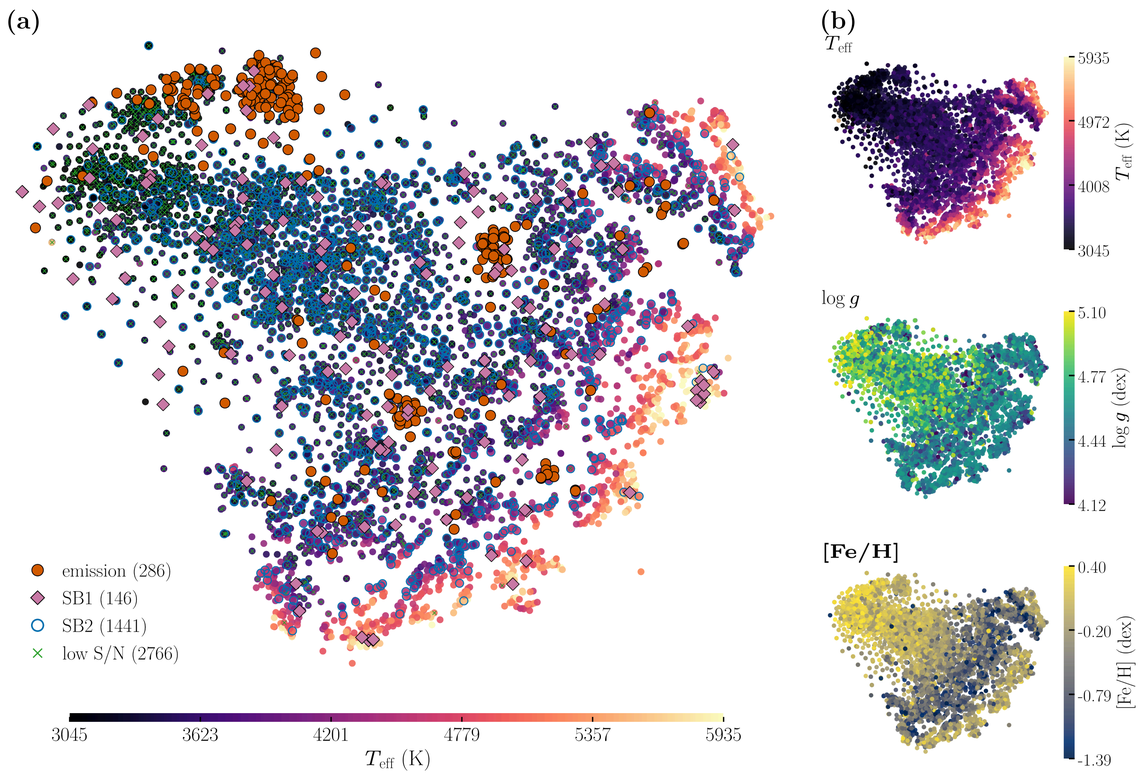


Figure 2 — Cuts and catalogue flags on the same map (grey: hidden stars). **(a)** Top row, cuts: $T_{\rm eff} \geq 4500$ K, S/N ≥ 10 in 4 CCDs, and both; no re-projection. **(b)** Bottom row, `flag_sp`: 0 = no warning; bit 1: Hα emission; 2: CCD missing; 4, 8: SB1, SB2; 2048: low S/N (52% of mapped stars). Flags are labels, not classes.

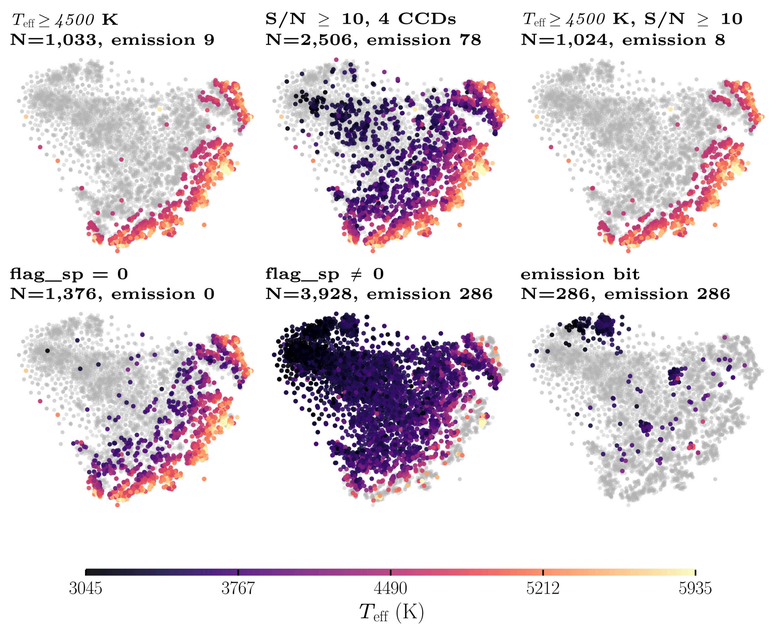


Figure 3 — Emission-flagged stars (286). **(a)** HDBSCAN finds five candidate subgroups: S0 (138 stars), S1 (12), S2 (54), S3 (12), S4 (34); 36 stay unassigned (dark grey; light grey: stars without the flag). **(b)** $T_{\rm eff}$: S0 concentrates cool M dwarfs (∼3400 K). **(c, d)** Kiel diagrams ($\log g$ vs $T_{\rm eff}$) over the 5,304 mapped stars (grey): S0 follows the solar-metallicity dwarf sequence, while S1 and S2 reach metal-poor regimes ($[Fe/H] < -0.5$). The subgroups are candidates; their nature is under investigation.

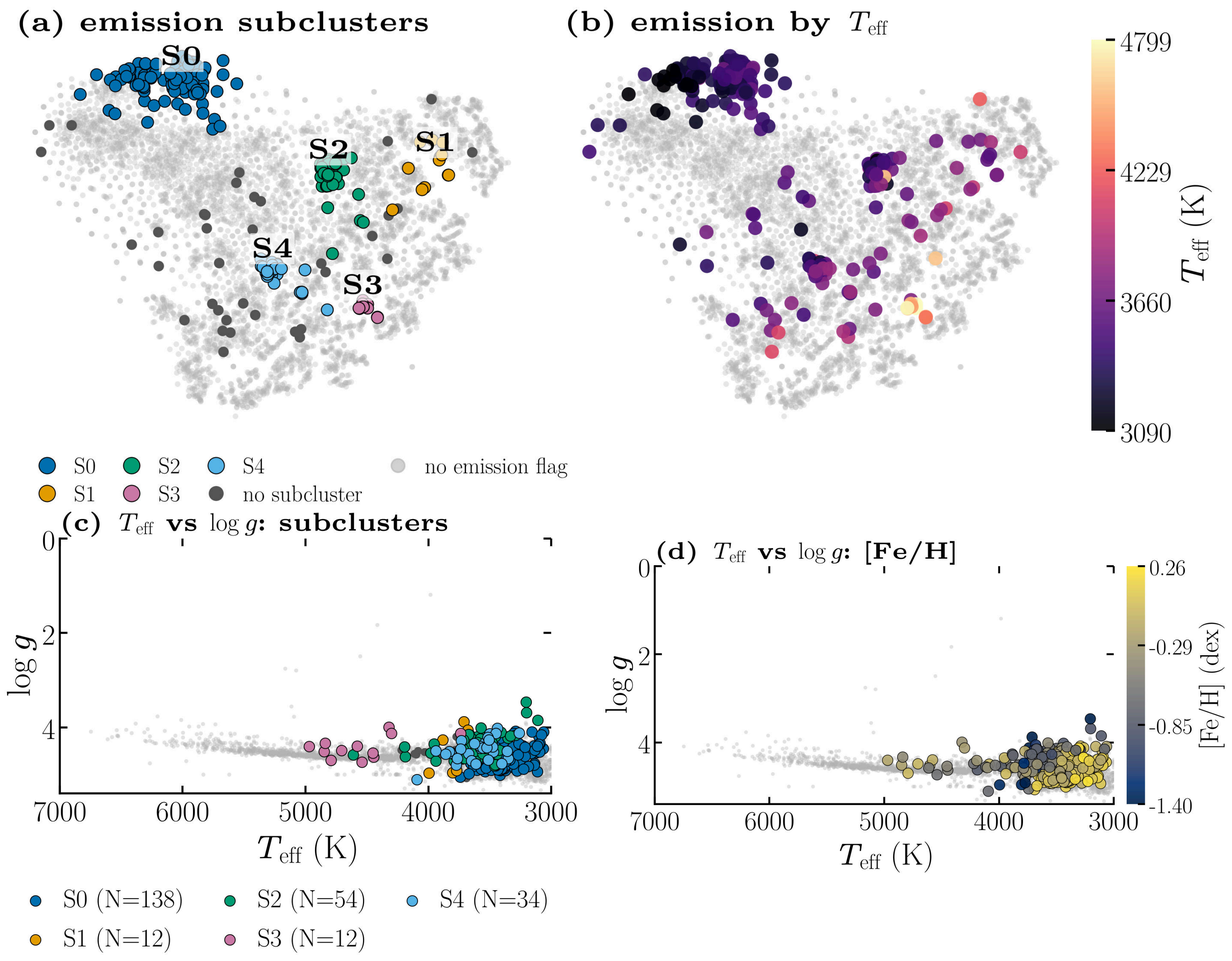


Figure 4 — Other settings: *t*-SNE (perplexity 30, 50) and UMAP (15, 30 neighbours) with a shared $T_{\rm eff}$ scale. Higher values raise the global distance correlation (up to 0.49); lower ones resolve finer substructure. Emission stars (orange) stay grouped in all four.

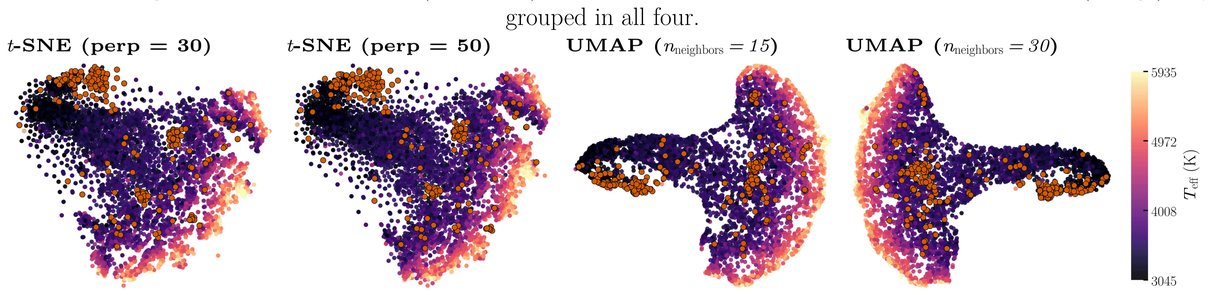
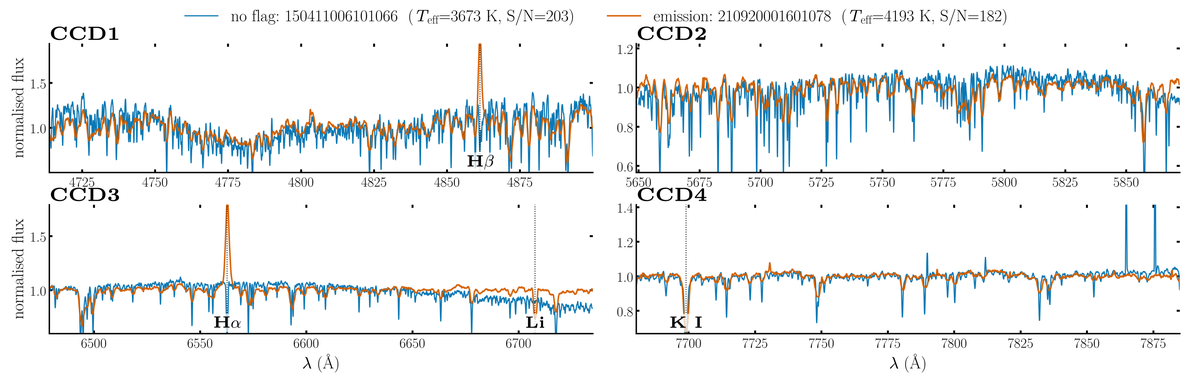


Figure 5 — Spectra across the four GALAH CCD channels: emission candidate (orange) vs. quiet dwarf (blue). A flag is only a catalogue label; the spectrum shows what the star is doing. A real emitter has Hα (and Hβ) filled in or in emission, which a quiet dwarf of similar $T_{\rm eff}$ lacks, separating genuine chromospheric activity from a spurious flag.



4. Key Findings

- $T_{\rm eff}$ recovered without labels in every setting.
- Cuts hide stars on one map (no re-projection); 52% carry the catalogue low-S/N flag, separate from our S/N ≥ 10 cut.
- Candidates, not classes: HDBSCAN settings and cuts not yet swept.
- Emitters stay grouped in all settings while local and global structure vary; the flagged example shows a real Hα peak.
- Next: all GALAH DR4 (905,415 spectra).

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